

DIY Lab: Speed of Sound Dependence on Temperature and Pressure

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PHYS 121 – Integrated Science

Dec 3, 2025

How does the speed of sound depend on temperature and pressure?

Why we investigated this:

- Previous lab: speed of sound at **fixed** conditions
- Real environments: temperature and pressure vary widely
- Goal: understand how these factors affect sound propagation

Project Goals:

- **Temperature dependence:** 20–80°C
- **Pressure dependence:** 100% to 0% atmospheric

What We Expected

Speed of sound in air:

$$v = 331.5 \sqrt{\left(1 + \frac{t}{273.15}\right) \left(1 + 0.16 \frac{rp_s}{p}\right)}$$

Ideal Gas Prediction:

- **Temperature:** dominant effect ($v \propto \sqrt{T}$)
- **Humidity:** minor correction factor

$$v = \sqrt{\gamma RT/M}$$

→ **Independent of pressure**

Expected behavior: Increasing $\sqrt{T}$ dependence; constant with pressure.

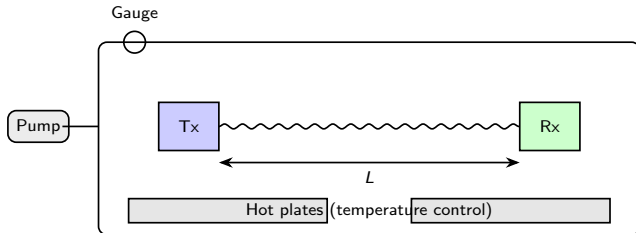
How We Measured It

Measurement approach:

- Traveling-wave phase comparison (Lab C)
- Ultrasonic transducers (40 kHz)
- Oscilloscope for phase shift detection
- Vernier caliper (0.01 mm) for displacement

Environmental Control:

- **Temperature:** hotspot-controlled chamber (20–80°C)
- **Pressure:** vacuum chamber (100–0%)



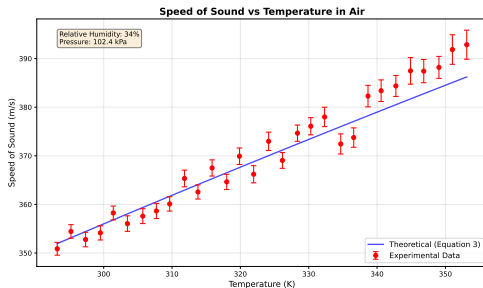
Results: Temperature Dependence

Observations:

- 30 data points (20–80°C)
- Speed increased from 351 m/s $\rightarrow$ 391 m/s
- Matches $\sqrt{T}$ prediction

Data Quality:

- Avg. error: **0.8%**
- Fit quality: $R^2 > 0.999$



Results: Pressure Dependence

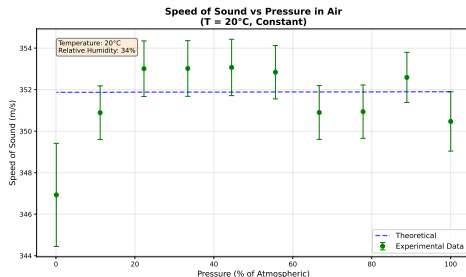
Observations:

- 10 measurements (100%→0.1% atm)
- Temperature fixed at 20°C
- Speed remained **nearly constant**

Theory: For ideal gas, speed depends only on temperature.

Error context:

- Variations $< 2\%$ within uncertainty
- Larger fluctuations at $< 1\%$ pressure



How Good Were Our Results?

Main error sources:

- **Measurement:** caliper resolution, frequency accuracy
- **Environment:** temperature drift, pressure instability
- **Detection:** phase difference reading

Error Trends:

- Low T: 0.4–0.5%
- High T: 1.0–1.5%
- Very low P: $\sim 1.4\%$

Experimental Challenges:

- Temperature stability of hot plates
- Vacuum chamber limitations at very low pressure
- Long thermal equilibration time

Overall uncertainty: 1.5 m/s – 2.0 m/s (0.4–1.5%)

What We Learned

- ① **Temperature dependence confirmed:** $v \propto \sqrt{T}$
- ② **Pressure independence verified:** constant within error
- ③ **High precision:** average deviation 0.8%
- ④ **Physical insight:** sound speed depends on molecular motion

Conclusion: Strong alignment between theory and experiment across a wide parameter range.

Limitations & Future Work

Limitations:

- Equipment-limited temperature range
- Difficult to maintain $< 1\%$ pressure
- Ideal-gas model breaks at high temperatures

Future Work:

- Explore humidity effects
- Test other gases (CO_2 , He)
- Improve temperature uniformity

Our Key Finding

Sound speed in air **increases with temperature** ($v \propto \sqrt{T}$)

but is **independent of pressure** in the ideal gas regime.

Thank you!

Questions?

- Lab instructors: Kai Wang, Chen Xi
- Course instructor: Prof. Changcheng Zheng
- Prof. Xiawa Wang's Lab (equipment & support)